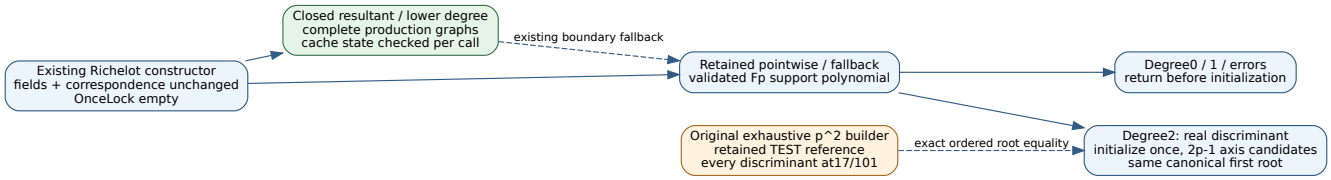
# Canonical roots without eager quadratic-field enumeration

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The Richelot reference evaluator now defers its root table until a degree2 support actually needs it. Its typed input is a validated polynomial over Fp, so every queried discriminant is real. Enumerating the two coordinate axes preserves the original exhaustive table’s first root for every query. The original builder remains a test reference; no field, polynomial, Cantor, correspondence or infinity formula is replaced. Native control01 and strictClippy pass, PCC3f015c. Complete changed guard02 now passes, PCC3c6362. [pcc:unverified]




## Why the first root is unchanged

Write Fp²=Fp[s]/(s²-n), n nonsquare and p odd. The square of a+b s has second coefficient2ab. A real square therefore forces a=0 or b=0. Filtering the old lexicographic enumeration to those axes gives all(0,b) first, then(a,0) for a=1..p-1. This is exactly the new order, with the origin once. Every matching candidate survives, so the first matching root is unchanged. [pcc:unverified]

RootTable.first_match_filter/PCC1da835 proves that ordered-search implication with an explicit match-implies-keep hypothesis. The local kernel/axiom gate audits one theorem, propext only, zero trusted native. Actual finite fields, basis/no-zero-divisors and concrete grid equality remain written bridges. ROOT_TABLE.md/PCC4ce55c is identical to upstream ROOT_TABLE_PROOF.md/718425. Formalization is partial. [pcc:unverified]

## Static enumeration counts, not timings

| Prime | Original candidates | Axis candidates |
|---|---:|---:|
| 17 | 289 | 33 |
| 101 | 10,201 | 201 |

The counts are p² and2p-1 candidate squares. The restricted table has p real keys, while the original table also stores nonreal square keys that this typed API never queries. This does not implement a general Fp² square-root table or establish a wall-clock, memory or DLP speed ratio. [pcc:unverified]

# Whole-map controls and evidence boundaries

## Lazy initialization is part of the tested behavior

The immutable map owns std::sync::OnceLock. Construction leaves it empty. Degree0/1 returns and zero/unsupported-degree errors occur before its initializer. The first degree2 pointwise or fallback call initializes the restricted table; later calls reuse it. The two returned polynomial roots have the same order and sign as before, including a repeated zero root. All original pointwise and geometric fallback formulas remain. [pcc:unverified]

Native01 compares EVERY base discriminant at17/101 against the original exhaustive $p^2$ builder. It checks exact canonical first roots, both returned polynomial zeros, every stored key/root and low-degree/error cache state. Three source01 snapshots and their hash manifest are frozen, PCC3f015c. The accepted field/model scope is unchanged; no larger-prime catalog acceptance is inferred from the general written argument. [pcc:unverified]

## Complete graph acceptance

The full R0/R1/R2 graph tests use separate fresh production/reference instances. Each closed production call must leave its own cache empty; actual pointwise quadratic references initialize theirs. Both directions retain complete domains, exact output equality, dual2, bijections, paired additions, descent, wrong-model rejection and geometric boundaries. All Phi/Psi/generic/naive/source controls remain in the whole guard. Complete changed check.sh02 passes25native/82reference unit plus five integrations, one pre-existing ignored benchmark, all guards/fmt/strictClippy. All six domains have239 lower-degree/12078 quadratic/zero fallback calls; every no-cache/reference-initialization assertion passes. Exact final three hashes match source01. Complete100487-byte raw→standard-xz11812bytes, raw/archive SHA256 and exact roundtrip are pinned by PCC3c6362. [pcc:unverified]

## Negative findings and retained failures

The initial proposal to remove lower-degree Richelot root finding was a recorded no-op: those fibre and infinity formulas are already algebraic. Upstream source audit325922 retains the exact old sources. This work instead changes table setup and queried-key scope. The first Lean parse used reserved token matches as a binder; source/error01 are retained, corrected hit in02 passes. No test, validator, frozen subject or ledger history is weakened. [pcc:unverified]

## Limits and next step

No stage, pipeline, index-calculus or rho advantage is measured. Matched cold protocols must include constructor/setup, conversion, transfer, solver, recovery and verified answers after S3/livebudget admission. Current locked atlas, frozen-ledger forks and parent CI findings remain separate blockers. Parent1496 exact65a focusedCI43136703928 succeeds25/81+five; its full archive/PCCe2187d remains separate. Own final-head CI/review and bottom-up merge requirements remain owed. [pcc:unverified]

Next CT27 prepares arbitrary balanced-class replay with the existing second Cantor engine, including the explicit infinity correction missing from the fixed49-class corpora. It is prospective. Independent isogeny evaluation/ kernel membership, larger-prime catalog support and full geometric Lean remain open. Reproduce using source01, the original exhaustive test builder, ROOT_TABLE.md and the ordinary check.sh; raw correctness receipts are in `evidence/ct26-*`. [pcc:unverified]